

Lecture : 5

Basic Components of a Computer System

Input Unit · Output Unit · ALU · Control Unit · CPU

Learning Outcomes

- Recall the Input → Process → Output cycle from Lecture 1 and identify which physical unit performs each stage.
- Define the Input Unit and list common input devices with a real-life example each.
- Define the Output Unit and list common output devices with a real-life example each.
- Explain the meaning, function and importance of the Central Processing Unit (CPU) as the “brain” of a computer.
- Describe the Arithmetic Logic Unit (ALU) and distinguish between its arithmetic and logical operations.
- Describe the Control Unit (CU) and explain the Fetch → Decode → Execute cycle it manages.
- Draw and label a complete block diagram of a computer system, showing how the Input Unit, CPU, Output Unit, Memory and Storage are connected by the system bus.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations, diagrams, and examples.

1 Recap — Opening Up the “Process” Box

In Lecture 1, we defined a computer as an electronic device that works on the **Input → Process → Output (IPO) cycle**: data goes in, the CPU processes it, and information comes out. That lecture treated “Process” as a single box — something that simply happens inside the machine.

Today we open up that middle box. A computer is not one single part; it is a small team of specialised components, each with one clear job, working together in a fixed sequence for every single instruction. This lecture introduces that team: the **Input Unit**, the **Output Unit**, and the **Central Processing Unit (CPU)** — which is itself made up of the **Arithmetic Logic Unit (ALU)**, the **Control Unit (CU)**, and a small set of **Registers**.

Important Point for Exams

The IPO cycle from Lecture 1 and the components in this lecture are the **SAME** idea at two levels of detail. Input Unit performs the “Input” stage; the CPU (ALU + Control Unit) performs the “Process” stage; the Output Unit performs the “Output” stage. Examiners often ask you to connect the two lectures in one answer.

2 The Basic Components of a Computer System

Every computer, regardless of its size or purpose, is built from five basic components. Three of them form the path data takes through the machine (Input Unit → CPU → Output Unit); the other two — Memory and Storage — hold the program and data the CPU is working with. Fig 5.1 shows the main flow.

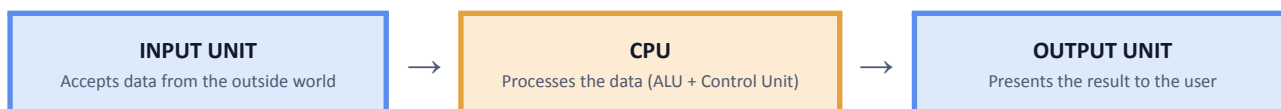


Fig 5.1 — The main flow of a computer system (compare with Fig 1.2, Lecture 1)

Component	One-line function
Input Unit	Accepts data and instructions from outside and converts them into binary form.
Central Processing Unit (CPU)	Processes data — performs calculations, comparisons and coordinates every other component.
Output Unit	Converts the CPU's binary result back into a form a human can understand.
Memory Unit (RAM)	Temporarily holds the program and data the CPU is currently using.
Secondary Storage	Permanently stores programs and data, even when the power is switched off.

3 The Input Unit

The Input Unit is the part of a computer system that accepts data and instructions from the outside world — usually from a human being — and converts them into a machine-readable, binary form that the CPU can process.

Functions of the Input Unit

- Accepts raw data and instructions from the user or from another device.

- Converts this data into binary digits (0s and 1s) — the only language the CPU understands.
- Supplies the converted data to the memory unit or directly to the CPU, along the system bus, for processing.

Common Input Devices

Device	What it converts to input	Real-life example
Keyboard	Key presses → character codes	Typing an Aadhaar enrolment form at a Common Service Centre (CSC)
Mouse	Movement and clicks → cursor position and commands	Clicking “Submit” on an online examination portal
Scanner	A printed image or document → a digital image	Scanning an old land-record register at a tehsil office
Microphone	Sound waves → digital audio data	Giving a voice command to a smartphone assistant
Barcode / QR Scanner	A printed code → a product or payment ID	Scanning a QR code to make a UPI payment at a village shop
Biometric (Fingerprint) Scanner	A fingerprint pattern → a digital template	Aadhaar-based biometric verification at a bank

Important Point for Exams

Memorise this one-sentence definition: the Input Unit converts human-understandable data into machine-understandable (binary) data. Examiners frequently ask you to name any four or five input devices — always add one real-life example next to each device for full marks.

4 The Output Unit

The **Output Unit** is the part of a computer system that takes the processed result from the CPU — which exists internally as binary data — and converts it into a form a human being can understand and use, such as text, images, sound, or a printed page. It performs exactly the reverse job of the Input Unit.

Common Output Devices

Device	What it converts binary into	Real-life example
Monitor (VDU)	A visual display on screen	Viewing examination results on a computer at a cyber café
Printer	A printed page	Printing a mark-sheet or a ration card at a CSC
Speaker	Sound	A voice assistant reading out today's weather forecast

Device	What it converts binary into	Real-life example
Plotter	A large-format printed drawing	Printing an engineering site map or building plan
Projector	A large projected image	Displaying a lecture slide on a classroom screen

Comparison: Input Unit vs Output Unit

Point	Input Unit	Output Unit
Direction of conversion	Human-understandable form → binary	Binary → human-understandable form
Position in the IPO cycle	Performs the “Input” stage	Performs the “Output” stage
Data flow	Data flows INTO the CPU/memory	Data flows OUT of the CPU/memory
Examples	Keyboard, Mouse, Scanner, Microphone	Monitor, Printer, Speaker, Plotter

5 The Central Processing Unit (CPU) — The Brain of the Computer

The Central Processing Unit (CPU) is the most important component of a computer system — often called its “brain” — because it is responsible for fetching, interpreting and executing every single instruction in a program, and for coordinating the activity of every other component.

The CPU is not one single block — it is itself built from three smaller parts working together: the Control Unit, the Arithmetic Logic Unit (ALU), and a small set of Registers. Fig 5.2 shows how these fit inside the CPU.

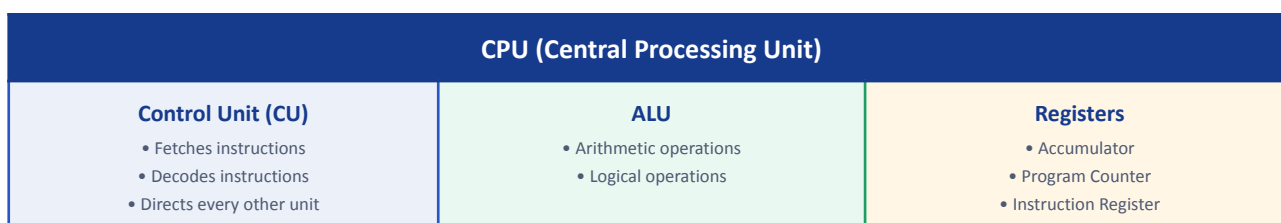


Fig 5.2 — Inside the CPU: Control Unit, ALU, and Registers

An Everyday Analogy — the Hostel Mess Kitchen

Picture the CPU as the kitchen of a large college hostel mess preparing today's recipe. The Control Unit is the head cook, who reads the recipe one step at a time and tells each helper exactly what to do and when — the head cook never chops a single vegetable. The ALU is the helper who actually does the chopping, mixing, and measuring — the real arithmetic and logical work. The Registers are the small steel bowls kept right next to the cook, holding only the one or two ingredients needed for the very next step — not the whole pantry, just what's needed right now.

6 The Arithmetic Logic Unit (ALU)

The Arithmetic Logic Unit (ALU) is the part of the CPU that actually performs every arithmetic calculation (such as addition, subtraction, multiplication and division) and every logical operation (such as AND, OR, NOT, and comparisons like greater-than or equal-to).

At its simplest, the ALU is built from arrangements of logic gates. For example, an XOR gate and an AND gate wired together form a half adder, which can add two single binary digits — in binary, $1 + 1 = 10$, just as $9 + 1 = 10$ in decimal. Millions of such gate arrangements, wired together, allow the ALU to add large numbers, compare values, and make decisions, all within a tiny fraction of a second.

Arithmetic Operations vs Logical Operations

Type	Examples of operations	What it is used for
Arithmetic operations	Addition (+), Subtraction (−), Multiplication (×), Division (÷)	Calculating totals, interest, averages, bills
Logical operations	AND, OR, NOT, and comparisons (>, <, =)	Making decisions and comparing two values

Real-Life Example

When a bank's computer checks whether a customer's account balance (₹5,000) is greater than or equal to the withdrawal amount requested at an ATM (₹2,000) before dispensing cash, this comparison is a logical operation carried out by the ALU — not a decision made by the machine “thinking” about it.

7 The Control Unit (CU)

The Control Unit (CU) is the part of the CPU that performs no calculations itself. Instead, it directs and coordinates the activities of every other part of the computer, by generating precisely-timed electronic control signals that tell each component exactly when to act.

The Control Unit repeatedly performs a three-step cycle for every single instruction in a program:

- **FETCH:** The next instruction is retrieved from memory.
- **DECODE:** The Control Unit works out what the instruction means, and which other parts of the CPU need to be involved.
- **EXECUTE:** The instruction is carried out — often by directing the ALU, or by moving data between registers and memory.

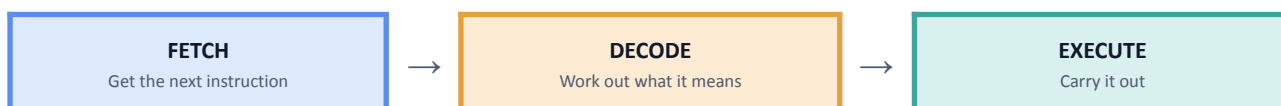


Fig 5.3 — The Fetch → Decode → Execute cycle. This repeats continuously, billions of times per second, for as long as the computer is switched on.

Common Doubt — Does the Control Unit Calculate?

This is a favourite examiner's trick question. The Control Unit never performs a calculation itself — continuing our mess-kitchen analogy, it only ever reads out the next line of the recipe and tells the correct helper to act. All arithmetic and logical calculation is done by the ALU. Confusing the two is one of the most common mistakes students make in this topic.

Registers — the CPU's Own Small, Fast Memory

Registers are extremely small, extremely fast storage locations built directly inside the CPU, used to hold only the one or two pieces of data needed for the instruction currently being executed — not the whole program. Two commonly named registers are the Accumulator, which holds the result of the calculation currently in progress, and the Program Counter (also called the Instruction Pointer), which keeps track of which instruction comes next.

Comparison: ALU vs Control Unit

Point	ALU	Control Unit
Main function	Performs arithmetic and logical operations	Directs and coordinates all other components
Does it calculate?	Yes — this is its entire job	No — it never calculates
Does it generate control signals?	No	Yes — this is its entire job
Everyday analogy	The kitchen helper who chops and measures	The head cook who reads the recipe aloud

8 Putting It All Together: The Complete Computer System

All five basic components — Input Unit, CPU, Output Unit, Memory Unit, and Secondary Storage — are connected by a set of electronic pathways called the system bus, along which data, memory addresses, and control signals travel between components.

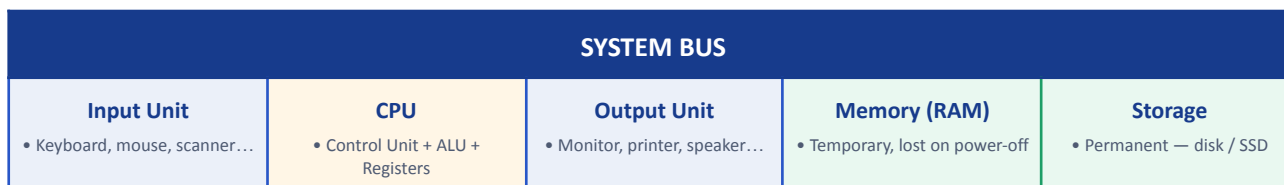


Fig 5.4 — The complete computer system: every component connects to the shared system bus

Worked Example — Tracing a Cash Withdrawal at an ATM

Step	Component involved	What happens
1	Input Unit	The customer inserts the card and types the PIN and amount on the keypad.
2	Control Unit	Fetches and decodes the “verify PIN” instruction from the program held in memory.
3	ALU	Compares the entered PIN against the stored PIN, and compares the account balance against the requested amount (logical/comparison operations).
4	Memory / Storage	The account balance is read from, and then updated in, the bank's stored records.
5	Output Unit	Cash is dispensed and a printed receipt or on-screen message is produced.

9 Fill in the Blanks

- 1. The _____ Unit accepts data from the outside world and converts it into binary form.
- 2. The _____ Unit converts the CPU's binary result into a form a human can understand.
- 3. The CPU is made up of the Control Unit, the _____, and Registers.
- 4. The _____ performs all arithmetic and logical operations inside the CPU.
- 5. The _____ directs and coordinates the activities of every other component, but never calculates.
- 6. The three steps of the cycle the Control Unit repeats for every instruction are Fetch, _____, and Execute.
- 7. The register that keeps track of which instruction comes next is called the _____.
- 8. The register that holds the result of the calculation currently in progress is called the _____.
- 9. The _____ is the set of electronic pathways that connects the Input Unit, CPU, Output Unit, Memory, and Storage.
- 10. Comparing whether a bank balance is greater than or equal to a withdrawal amount is an example of a _____ operation.

Answers

1. Input | 2. Output | 3. ALU (Arithmetic Logic Unit) | 4. ALU | 5. Control Unit | 6. Decode | 7. Program Counter / Instruction Pointer | 8. Accumulator | 9. System bus | 10. Logical

10 True or False

- 1. The Input Unit converts binary data into a form a human being can understand.
- 2. The CPU is made up of the Control Unit, the ALU, and Registers.
- 3. The Arithmetic Logic Unit (ALU) performs both arithmetic and logical operations.
- 4. The Control Unit performs calculations, just like the ALU.
- 5. Registers are large, permanent storage areas located outside the CPU.
- 6. The Fetch → Decode → Execute cycle repeats continuously while the computer is switched on.
- 7. The system bus connects the Input Unit, CPU, Output Unit, Memory, and Storage.
- 8. A printer is an example of an input device.

Answers with Explanation

1. False. The Input Unit does the opposite — it converts human-understandable data into binary. The Output Unit converts binary into a human-understandable form.

2. True. These three parts together make up the CPU, as shown in Fig 5.2.

3. True. Arithmetic operations (like addition) and logical operations (like comparisons) are both carried out by the ALU.

4. False. The Control Unit never calculates — it only directs and coordinates. All calculation is done by the ALU.

5. False. Registers are extremely small and are built directly inside the CPU, not outside it, and they hold only the data needed for the current instruction.

6. True. This cycle repeats billions of times per second for every instruction in every running program.

7. True. The system bus is the shared set of pathways connecting all five basic components, as shown in Fig 5.4.

8. False. A printer is an output device — it presents the CPU's result to the user. Examples of input devices include the keyboard, mouse, and scanner.

11 Short Answer Questions (2–3 Marks)

- Define the Input Unit and give two examples of input devices. (2 marks)
- Define the Output Unit and give two examples of output devices. (2 marks)
- What is the ALU? What two kinds of operations does it perform? (2 marks)
- What is the function of the Control Unit? (2 marks)
- Differentiate between the ALU and the Control Unit. (3 marks)
- Explain the Fetch → Decode → Execute cycle in brief. (3 marks)
- What are registers? Name any two registers and their function. (3 marks)

12 Long Answer Model Answers (5, 10 & 15 Marks)

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

Q1 (5 Marks): Explain the Input Unit and Output Unit of a computer with suitable examples.

Every computer system needs a way to receive data from the outside world and a way to present its results back to the user. These two jobs are handled by the Input Unit and the Output Unit respectively.

- **Input Unit:** The Input Unit accepts data and instructions from a human being or another device and converts them into binary form, the only language the CPU can process. For example, when a student types an answer into an online examination portal using a keyboard, the Input Unit converts each key press into a character code that the computer can process.
- **Output Unit:** The Output Unit performs the reverse job — it converts the CPU's internal binary result into a form a human can understand, such as text, an image, sound, or a printed page. For example, when a Common Service Centre prints a student's mark-sheet, the printer (an output device) converts the computer's binary result into printed text on paper.

Together, the Input Unit and the Output Unit form the two ends of the Input → Process → Output cycle studied in Lecture 1, allowing a human being to communicate with, and receive results from, the CPU.

Q2 (10 Marks): Explain the Central Processing Unit (CPU). Describe the functions of the ALU and Control Unit in detail, with a diagram.

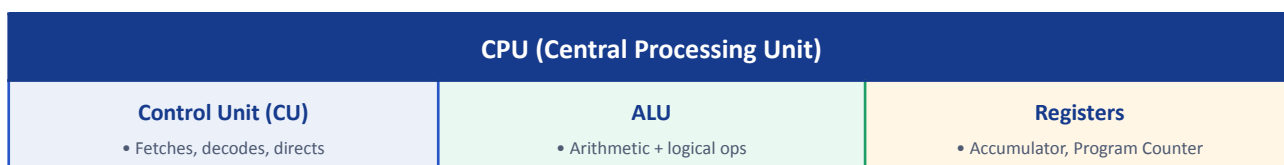
The Central Processing Unit (CPU) is the most important component of a computer system, often called its “brain”, because it is responsible for fetching, interpreting and executing every instruction in a program, and for coordinating the activity of every other component. The CPU itself is made up of three smaller parts: the Control Unit, the Arithmetic Logic Unit (ALU), and a small set of Registers.

The Arithmetic Logic Unit (ALU): The ALU is the part of the CPU that actually performs every arithmetic operation, such as addition, subtraction, multiplication and division, and every logical operation, such as AND, OR, NOT, and

comparisons like greater-than or equal-to. For example, when a bank's computer checks whether a customer's balance is sufficient before dispensing cash from an ATM, this comparison is carried out by the ALU.

The Control Unit (CU): The Control Unit performs no calculations itself. Instead, it directs and coordinates every other part of the computer by generating precisely-timed control signals. It repeatedly carries out a three-step cycle for every instruction: FETCH, where the next instruction is retrieved from memory; DECODE, where the Control Unit works out what the instruction means; and EXECUTE, where the instruction is carried out, often by directing the ALU. This Fetch → Decode → Execute cycle repeats continuously, billions of times per second, for as long as the computer is switched on.

Registers: Registers are extremely small, extremely fast storage locations built directly inside the CPU, holding only the data needed for the instruction currently being executed. The Accumulator holds the result of the calculation in progress, while the Program Counter keeps track of which instruction comes next.



(A labelled CPU diagram, as in Fig 5.2 of these notes, should be drawn here in your answer sheet to earn full diagram marks.)

In conclusion, the CPU's power comes from the close cooperation of these three parts: the Control Unit directs, the ALU calculates, and the Registers hold exactly what is needed at each step — together allowing the CPU to execute millions of instructions every second.

Q3 (15 Marks): “The CPU is the brain of the computer, but a brain cannot work without a body.” Discuss this statement with reference to the Input Unit, Output Unit, CPU (ALU + Control Unit + Registers), Memory, Storage, and the system bus, using a complete labelled block diagram and real-life examples.

This statement captures an important truth about how a computer system works: the CPU is indeed the most powerful and important component, but it is completely useless on its own — it depends entirely on the Input Unit to supply it with data, the Output Unit to present its results, and Memory and Storage to hold the program and data it works on. This answer discusses each of these components and how they work together.

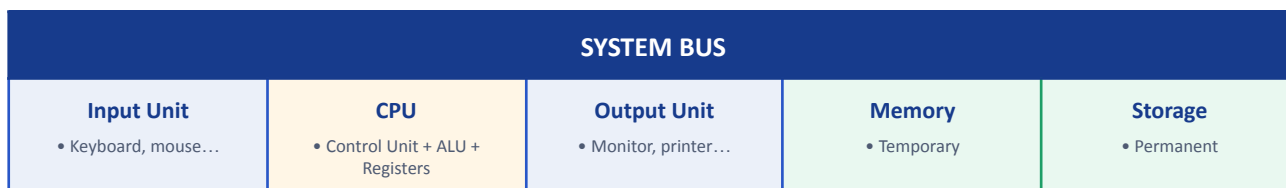
The Input Unit: The Input Unit accepts data and instructions from the outside world — typically from a human being, using devices such as a keyboard, mouse, scanner, or microphone — and converts them into binary form. Without the Input Unit, the CPU would have no data to process at all, no matter how fast or powerful it is.

The CPU — the “brain”: The CPU is made up of the Control Unit, the ALU, and Registers. The Control Unit directs and coordinates all activity by generating control signals and running the Fetch → Decode → Execute cycle; the ALU performs every arithmetic and logical operation, such as addition or comparing two values; and the Registers hold the small amount of data needed for the instruction currently being executed. This is genuinely the “brain” of the system — but, like a human brain, it cannot sense the world or act upon it without a body around it.

The Output Unit: The Output Unit converts the CPU's binary results into a form a human being can understand — text, images, sound, or a printed page — using devices such as a monitor, printer, or speaker. Without it, the CPU's work would remain locked inside the machine, invisible and useless to the user.

Memory and Storage: The Memory Unit (RAM) temporarily holds the program and data the CPU is currently using, while Secondary Storage — such as a hard disk or SSD — holds programs and data permanently, even when the power is switched off. The CPU constantly reads from and writes to memory as it executes instructions; without memory, it would have nowhere to fetch instructions from in the first place.

The System Bus — connecting the body together: All of these components are connected by the system bus, a set of electronic pathways along which data, addresses, and control signals travel between the Input Unit, the CPU, the Output Unit, Memory, and Storage.



(A complete labelled block diagram, as in Fig 5.4 of these notes, should be drawn here in your answer sheet to earn full diagram marks.)

Worked example: Consider a cash withdrawal at an ATM. The Input Unit accepts the card, PIN and amount; the Control Unit fetches and decodes the “verify PIN” instruction; the ALU compares the entered PIN and the requested amount against the stored records; Memory and Storage supply and update the account balance; and finally the Output Unit dispenses the cash and prints a receipt. Every single component had to play its part — the CPU alone could not have withdrawn a single rupee.

In conclusion, the statement is entirely accurate. The CPU is genuinely the “brain” of the computer because of the remarkable speed and precision of the Control Unit and the ALU. But exactly like a human brain, which needs eyes, ears, hands, and a voice to interact with the world, the CPU needs the Input Unit, the Output Unit, Memory, and Storage — all connected by the system bus — to form a complete, useful computer system.

13 Exam Tips

- Memory trick for the five components: I-C-O-M-S — Input, CPU, Output, Memory, Storage.
- Memory trick for the CPU: “ALU calculates, CU coordinates.” If a question is really about calculation, it's the ALU; if it's about directing or timing, it's the Control Unit.
- Always mention the Fetch → Decode → Execute cycle when explaining the Control Unit — examiners specifically look for this term.
- Always draw the relevant diagram (Fig 5.1, 5.2 or 5.4) whenever you define the CPU or the complete computer system — diagrams earn dedicated marks in most university marking schemes.
- For “Differentiate” questions, present your answer as a table (see Sections 4 and 7) — it is faster to write and easier for the examiner to award marks clearly.
- For 10 and 15 mark questions, structure your answer with a clear paragraph for each component, exactly like the model answers in Section 12 — examiners reward clear structure.
- Common Mistake: Do not say the Control Unit “calculates” or “processes data” — that is the ALU's job. The Control Unit only directs and coordinates.
- Common Mistake: Do not say “CPU” when you mean the whole “computer” — the CPU is only one component, however important.